

Statistical Hypothesis Testing

Customer Sales Across Operating Regions

Portfolio Project Report

This report documents a customer-level statistical analysis of sales across the company's operating regions. The analysis accounts for repeated observations from customers who purchased across multiple regions.

Project focus	Customer-level regional sales differences
Dataset	Superstore sales data used in the Jupyter Notebook
Primary outcome	Sales
Significance level	$\alpha = 0.05$
Main designs	Paired comparison and repeated-measures comparison
Notebook	Hypothesis_Testing.ipynb

Project Overview

The objective of the analysis is to determine whether customer-level sales differ across the company's operating regions. A key feature of the data is that customers are not necessarily restricted to one region. Because the same customer can contribute sales to multiple regions, treating all regional observations as independent would ignore the within-customer dependence.

The notebook therefore uses within-customer statistical designs. Two related questions are examined: first, whether East and West sales differ among customers who purchased in both regions; and second, whether sales differ across Central, East, South and West among customers who purchased in all four regions.

The analysis uses $\alpha = 0.05$ throughout.

Data Preparation

The notebook loads the Superstore dataset and checks the data before conducting the hypothesis tests. The dataset contains 21 columns, including Customer ID, Region, Sales, Quantity, Discount and Profit.

Data check	Notebook result
Missing values	0 missing values across the 21 variables
Duplicate rows	0 duplicated rows
Negative Sales values	0 observations
Unique customers represented	793
Customer-region combinations	2,501

The customer-region aggregation produces one sales total per customer-region combination. Of the 793 customers, 301 purchased in all four regions, 340 purchased in three regions, 125 purchased in two regions and 27 purchased in one region.

East vs West: Paired Analysis

The first research question asks: *For customers operating in both the East and West regions, is their mean sales different?*

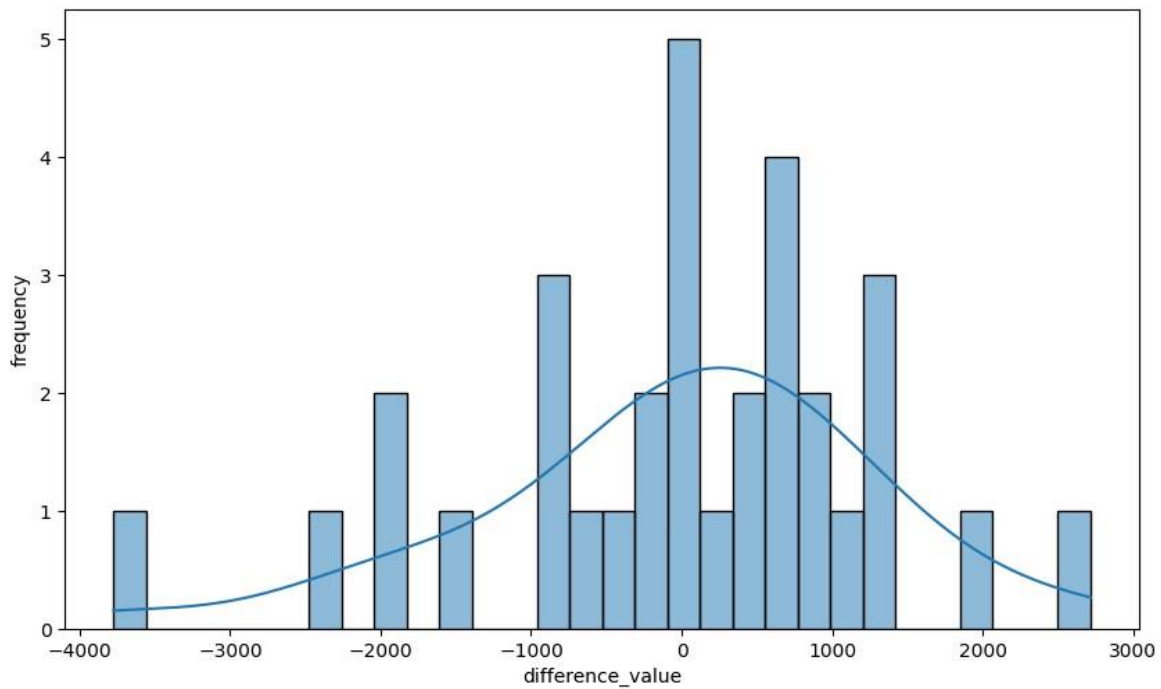
H₀: the mean East–West sales difference is 0.

H₁: the mean East–West sales difference is not 0.

The notebook identifies 32 customers with sales in both East and West. Sales are compared within the same customer, so each customer acts as their own paired comparison.

Descriptive results

Measure	Value
Customers	32
Mean difference (East – West)	–25.93
SD of differences	1,318.63
Median difference	47.81
Minimum difference	–3,773.91
Maximum difference	2,716.46



Distribution of customer-level East – West sales differences.

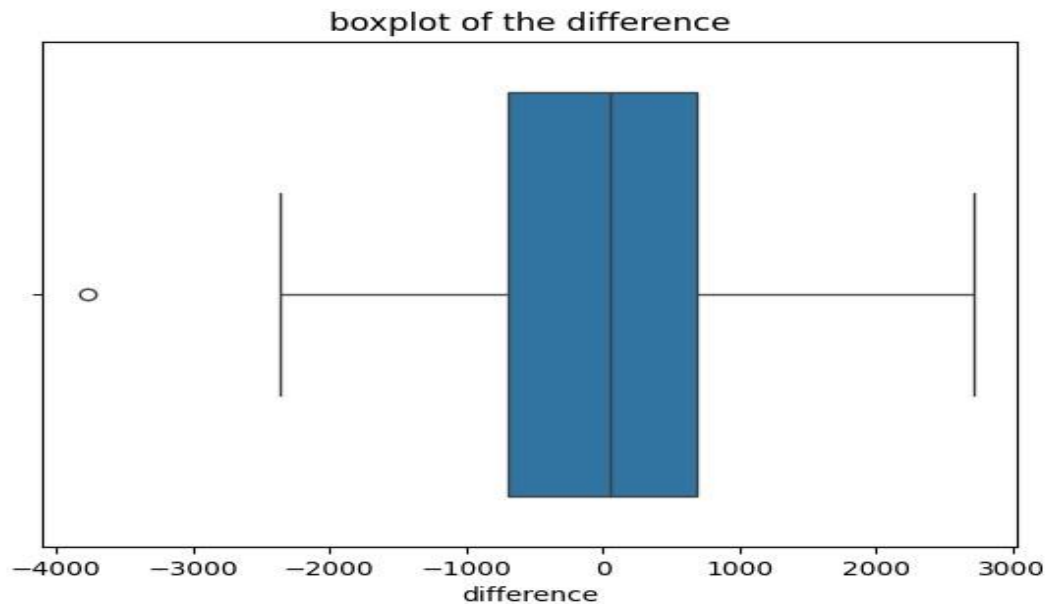
Primary paired t-test

Method	Statistic / effect	p-value	n
Paired t-test	$t = -0.1112$	0.9122	32
Cohen's d	$d = -0.0197$	—	32
Wilcoxon signed-rank	$W = 252$	0.8320	32
Sign-flip permutation	—	0.9123	32

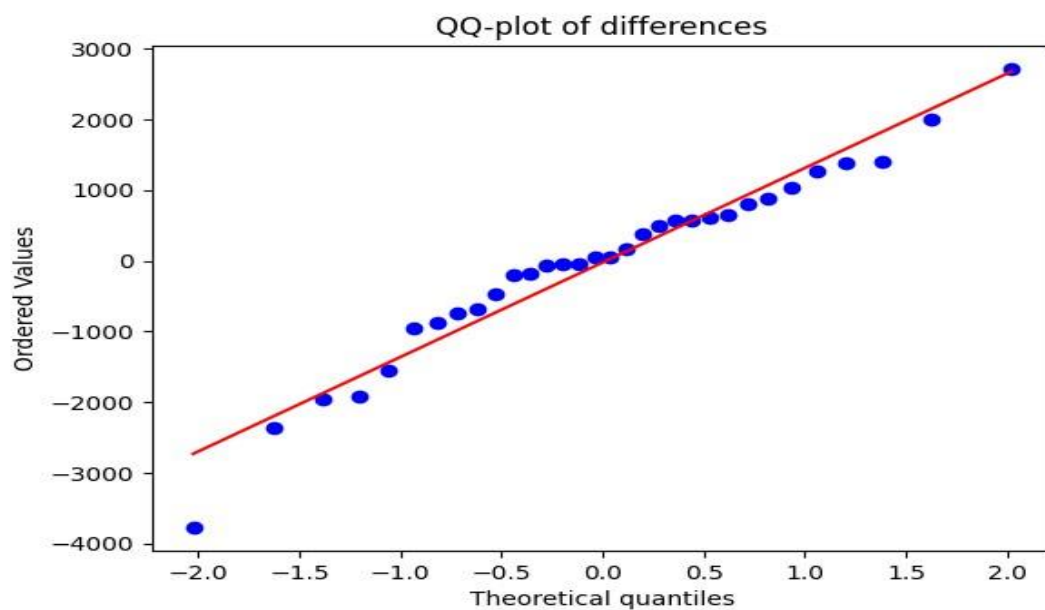
The paired t-test produced $t(31) = -0.1112$ and $p = 0.9122$. The 95% confidence interval for the mean difference was -501.34 to 449.49 , which includes zero. Cohen's d was -0.0197 . The Wilcoxon and sign-flip permutation checks also produced p-values above 0.05 (0.8320 and 0.9123 respectively).

Interpretation: Within the 32 customers who purchased in both regions, the notebook does not find statistically significant evidence of a difference in mean East versus West sales at the 5% significance level. The three inferential approaches point in the same direction.

East–West assumption and robustness visuals



Boxplot of East – West customer-level sales differences.



Q–Q plot used to assess the distribution of the paired differences.

Four-Region Repeated-Measures Analysis

The second research question asks: *For customers operating in all four regions, is their mean sales the same across Central, East, South and West?*

H₀: there is no difference in mean sales across the four regions.

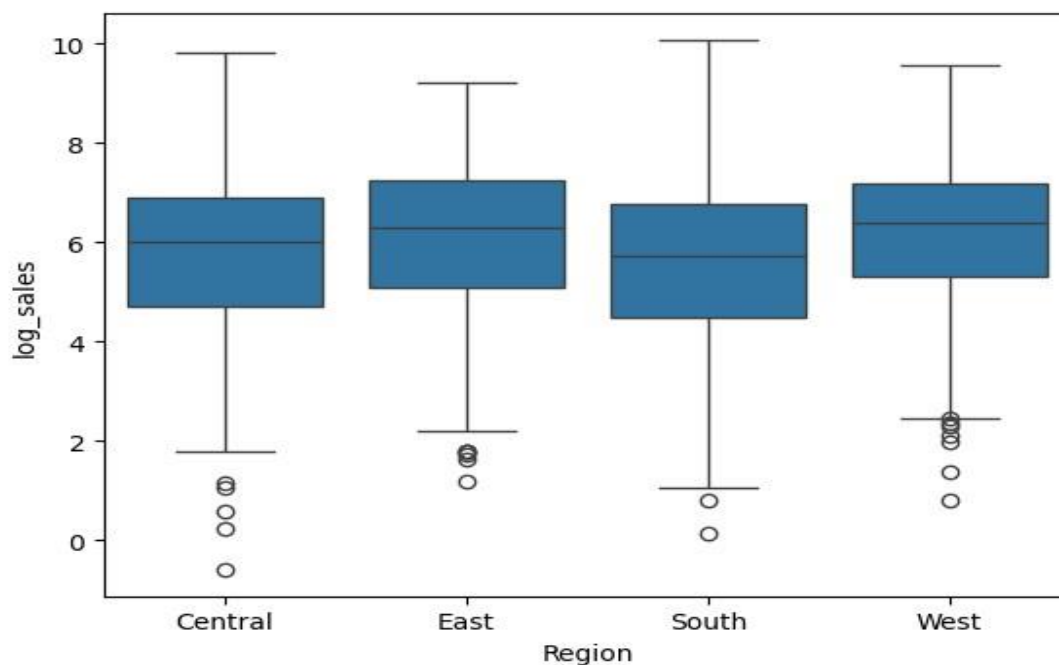
H₁: at least one regional mean differs.

The analysis is restricted to the 301 customers with observations in all four regions. Because the same customers appear in every region, a repeated-measures design is used.

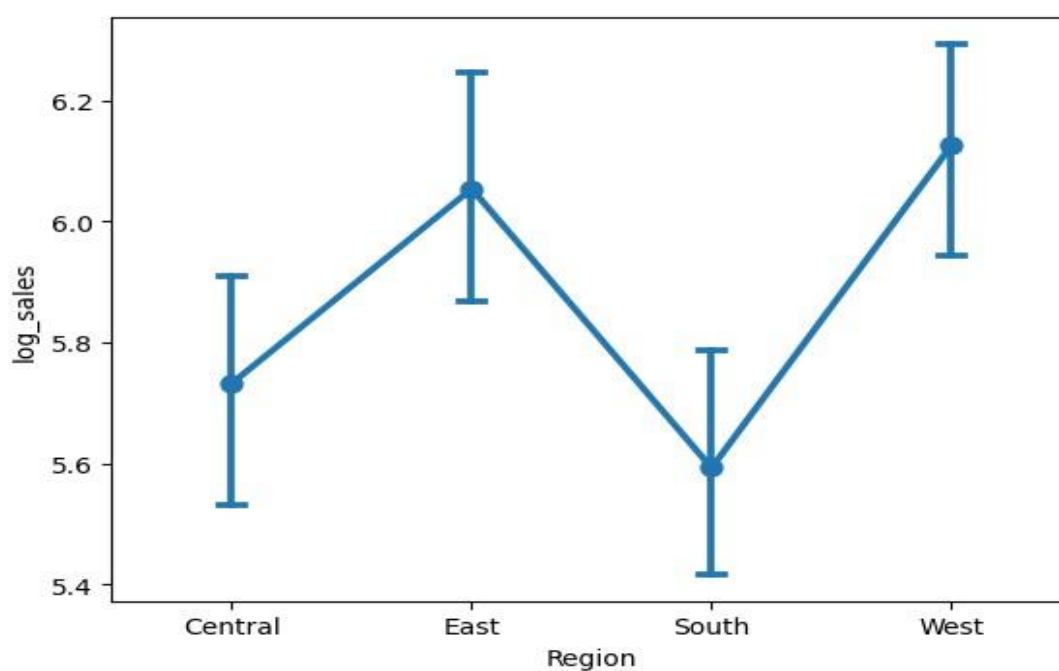
Raw-sales analysis and assumption checks

The notebook first examines the raw sales data. The raw-sales repeated-measures ANOVA produced $F = 2.6454$ with Greenhouse–Geisser-corrected $p = 0.049386$ and generalized eta-squared = 0.006657. Mauchly's test indicated that the sphericity assumption was not met ($W = 0.96197$, $p = 0.04099$).

The notebook then applies a log transformation to Sales because the sales values are highly right-skewed. The transformed outcome is $\log(\text{Sales})$, which reduces the influence of very large observations and provides a more suitable scale for the subsequent parametric analysis.



Log-transformed sales distribution by region.



Regional mean $\log(\text{Sales})$ with confidence intervals.

Log-transformed sales: descriptive statistics

Region	n	Mean log(Sales)	SD
Central	301	5.7322	1.6482
East	301	6.0539	1.5994
South	301	5.5925	1.6366
West	301	6.1258	1.4766

Normality and sphericity checks

The notebook evaluates the normality of the six within-customer pairwise differences. Central–East and Central–West differences were flagged as non-normal by the Shapiro–Wilk test ($p = 0.00747$ and $p = 0.00198$), while the remaining four pairwise differences were not flagged at the 5% level.

Mauchly's test on the log-transformed data also indicated a violation of sphericity: $W = 0.95755$, $\chi^2 = 12.9581$, $p = 0.02378$. The Greenhouse–Geisser correction was therefore used for the repeated-measures ANOVA.

Greenhouse–Geisser corrected repeated-measures ANOVA

Measure	Result
F statistic	7.6150
Uncorrected p-value	0.00005
Greenhouse–Geisser corrected p-value	0.00006
Generalized eta-squared	0.018951
Sample size	301 customers

The Greenhouse–Geisser corrected repeated-measures ANOVA gives $F = 7.6150$ and $p = 0.00006$, indicating a statistically significant regional effect on log-transformed sales at $\alpha = 0.05$. The generalized eta-squared reported by the notebook is 0.018951.

Holm-adjusted post-hoc comparisons

Comparison	Adjusted p	Cohen's d
Central vs East	0.049849	−0.198
Central vs South	0.594885	0.085
Central vs West	0.010482	−0.252
East vs South	0.004631	0.285
East vs West	0.594885	−0.047
South vs West	0.000400	−0.342

After Holm adjustment, the notebook identifies statistically significant pairwise differences for Central–East, Central–West, East–South and South–West. Central–South and East–West do not meet the 0.05 threshold after adjustment.

Within-customer permutation check

The notebook performs 10,000 within-customer permutations by shuffling the four regional sales values within each customer while preserving each customer's set of sales. The observed raw-sales F statistic was 2.6454 and the resulting permutation p-value was 0.0451.

Interpretation: The permutation analysis also falls below 0.05, although its p-value is much closer to the threshold than the Greenhouse–Geisser corrected analysis. Together with the repeated-measures ANOVA, the notebook provides evidence of regional differences among the 301 customers represented in all four regions.

1. Consolidated Results

Analysis	Method	n	Statistic	p-value
East vs West	Paired t-test	32	$t = -0.1112$	0.9122
East vs West	Wilcoxon	32	$W = 252$	0.8320
East vs West	Sign-flip permutation	32	—	0.9123
East vs West	Cohen's d	32	$d = -0.0197$	—
Four regions	GG-corrected RM-ANOVA	301	$F = 7.6150$	0.00006
Four regions	Within-customer permutation	301	—	0.0451

6. Key Findings

1. The customer-level East–West comparison does not provide statistically significant evidence of a mean sales difference among the 32 customers who purchased in both regions.
2. The four-region analysis uses a different customer population: 301 customers who purchased in all four regions. This repeated-measures analysis detects a regional effect on log-transformed sales.
3. The significant overall four-region result does not mean that every pair of regions differs. The Holm-adjusted post-hoc analysis identifies specific regional contrasts, while Central–South and East–West are not statistically significant after adjustment.
4. The East–West result is consistent across the paired t-test, Wilcoxon signed-rank test and sign-flip permutation test.

2. Limitations and Interpretation Notes

The East–West and four-region analyses are based on different subsets of customers. Therefore, their results should not be interpreted as contradictory tests on exactly the same population. The first analysis concerns 32 customers with both East and West sales; the second concerns 301 customers with sales in all four regions.

The analysis establishes statistical evidence of differences within the examined customer groups; it does not by itself establish why those differences occur. Factors such as customer purchasing behaviour, product mix, order volume and other business characteristics would require additional analysis before causal explanations could be made.

The repeated-measures analysis also involves assumption checks and corrections. Because Mauchly's test indicates that sphericity is violated, the Greenhouse–Geisser correction is used rather than relying on the uncorrected ANOVA p-value.

3. Conclusion

The analysis demonstrates why customer-level dependence matters when comparing regional sales. A simple independent-groups comparison would overlook the fact that many customers appear in multiple regions. By using paired and repeated-measures methods, the notebook compares regional sales within the same customers.

For the East–West comparison, the evidence does not indicate a statistically significant difference among the 32 customers who purchased in both regions. In contrast, the four-region repeated-measures analysis among 301 customers provides evidence of a regional effect on log-transformed sales, with several specific pairwise differences remaining significant after Holm adjustment.

4. Project Structure

Folder / file	Purpose
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Hypothesis_Testing.ipynb	Full analysis, code, tests and statistical outputs
Folder / file	Purpose
visualizations/	Selected charts used to communicate the analysis
results/	Tabulated statistical results and outputs
reports/Hypothesis_Testing_Report.pdf	Portfolio-ready written report of the project
README	Project overview, context and navigation

This report is intended to accompany the Jupyter Notebook and supporting visualization and results folders in the portfolio repository.